KAIZEN MATHS

Lesson starter

Permutations and Combinations

Permutations

Name

S1

Use permutation reasoning when order matters. Show how restrictions or repeated objects change the count.

A club has 11 members. How many ways can a president, secretary, and treasurer plus 1 other distinct roles be chosen?

S2

Use permutation reasoning when order matters. Show how restrictions or repeated objects change the count.

7 different books include a red book and a blue book. How many arrangements are possible if the red and blue books must be next to each other?

S3

Use permutation reasoning when order matters. Show how restrictions or repeated objects change the count.

How many distinct arrangements can be made from the letters of LEVEL?

S4

Use permutation reasoning when order matters. Show how restrictions or repeated objects change the count.

How many different circular arrangements are possible for 6 people around a round table?

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STA-70FE-L2 · S

Show your working

KAIZEN MATHS

Exit ticket

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EA1

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KAIZEN MATHS

Exit ticket

Permutations and Combinations

Permutations

Name

EB1

Use permutation reasoning when order matters. Show how restrictions or repeated objects change the count.

How many distinct arrangements can be made from the letters of SUCCESS?

Show your working

KAIZEN MATHS

Exit ticket

Permutations and Combinations

Permutations

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EA1

Use permutation reasoning when order matters. Show how restrictions or repeated objects change the count.

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EA2

Use permutation reasoning when order matters. Show how restrictions or repeated objects change the count.

6 different books include a red book and a blue book. How many arrangements are possible if the red and blue books must be next to each other?

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Show your working

KAIZEN MATHS

Exit ticket

Permutations and Combinations

Permutations

Name

EB1

Use permutation reasoning when order matters. Show how restrictions or repeated objects change the count.

How many distinct arrangements can be made from the letters of SUCCESS?

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Show your working

Teacher answers & working

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S1

Use permutation reasoning when order matters. Show how restrictions or repeated objects change the count.

A club has 11 members. How many ways can a president, secretary, and treasurer plus 1 other distinct roles be chosen?

Answer:

7,920

Working:

The roles are different, so order matters.

${}^{11}\mathrm{P}_4 = \frac{11!}{7!}$

${}^{11}\mathrm{P}_4 = 11 \times 10 \times 9 \times 8$

${}^{11}\mathrm{P}_4 = 7{,}920$

S2

Use permutation reasoning when order matters. Show how restrictions or repeated objects change the count.

7 different books include a red book and a blue book. How many arrangements are possible if the red and blue books must be next to each other?

Answer:

1,440

Working:

Treat the red and blue books as one block.

Number of objects to arrange = 6

Arrangements of the objects = 6! = 720

The two books inside the block can be ordered as red-blue or blue-red.

$720 \times 2 = 1{,}440$

S3

Use permutation reasoning when order matters. Show how restrictions or repeated objects change the count.

How many distinct arrangements can be made from the letters of LEVEL?

Answer:

30

Working:

Start with all arrangements, then divide by the arrangements of repeated letters.

Total arrangements before correcting repeats = 5!

Repeated-letter correction = 2! × 2!

Distinct arrangements = $\frac{5!}{2! \times 2!}$

Distinct arrangements = $\frac{120}{4}$ = 30

Teacher answers & working

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S4

Use permutation reasoning when order matters. Show how restrictions or repeated objects change the count.

How many different circular arrangements are possible for 6 people around a round table?

Answer:

120

Working:

For circular arrangements, rotations are considered the same.

Fix one person in position, then arrange the remaining people.

Number of arrangements = (6 - 1)!

$5! = 120$

EA1

Use permutation reasoning when order matters. Show how restrictions or repeated objects change the count.

A club has 10 members. How many ways can a president, secretary, and treasurer be chosen?

Answer:

720

Working:

The roles are different, so order matters.

$^{10}P_3 = \frac{10!}{7!}$

$^{10}P_3 = 10 \times 9 \times 8$

$^{10}P_3 = 720$

EA2

Use permutation reasoning when order matters. Show how restrictions or repeated objects change the count.

6 different books include a red book and a blue book. How many arrangements are possible if the red and blue books must be next to each other?

Answer:

240

Working:

Treat the red and blue books as one block.

Number of objects to arrange = 5

Arrangements of the objects = 5! = 120

The two books inside the block can be ordered as red-blue or blue-red.

$120 \times 2 = 240$

Teacher answers & working

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EB1

Use permutation reasoning when order matters. Show how restrictions or repeated objects change the count.

How many distinct arrangements can be made from the letters of SUCCESS?

Answer:

420

Working:

Start with all arrangements, then divide by the arrangements of repeated letters.

Total arrangements before correcting repeats = 7!

Repeated-letter correction = 3! × 2!

Distinct arrangements = $\frac{7!}{3!\times 2!}$

Distinct arrangements = $\frac{5{,}040}{12}$ = 420